10

Deconstructing the Bang

WHAT BANGED?

A common misconception is that the big bang provides a theory of cosmic origins. It doesn't. The big bang is a theory, partly described in the last two chapters, that delineates cosmic evolution from a split second after whatever happened to bring the universe into existence, but *it says nothing at all about time zero itself*. And since, according to the big bang theory, the bang is what is supposed to have happened at the beginning, the big bang leaves out the bang. It tells us nothing about what banged, why it banged, how it banged, or, frankly, whether it ever really banged at all.[1] In fact, if you think about it for a moment, you'll realize that the big bang presents us with quite a puzzle. At the huge densities of matter and energy characteristic of the universe's earliest moments, gravity was by far the dominant force. But gravity is an *attractive* force. It impels things to come together. So what could possibly be responsible for the *outward* force that drove space to expand? It would seem that some kind of powerful repulsive force must have played a critical role at the time of the bang, but which of nature's forces could that possibly be?

For many decades this most basic of all cosmological questions went unanswered. Then, in the 1980s, an old observation of Einstein's was resurrected in a sparkling new form, giving rise to what has become known as *inflationary cosmology.* And with this discovery, credit for the bang could finally be bestowed on the deserving force: *gravity.* It's surprising, but physicists realized that in just the right environment gravity can be repulsive, and, according to the theory, the necessary conditions prevailed during the earliest moments of cosmic history. For a time interval that would make a nanosecond seem an eternity, the early universe provided an arena in which gravity exerted its repulsive side with a vengeance, driving every region of space away from every other